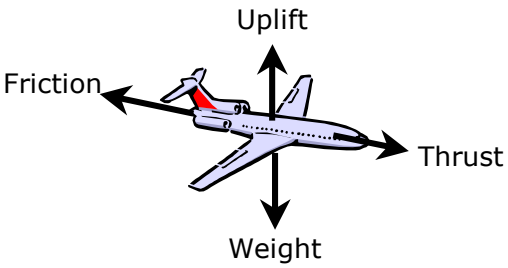
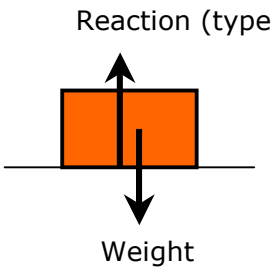
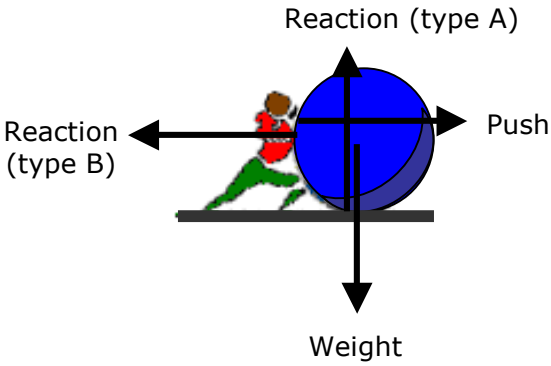
Types of Forces

There are many different types of forces. Some of them are shown in the pictures below.









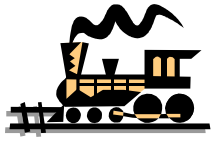
Task 1 - Force definitions

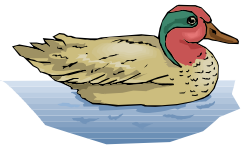
Match each type of force with its description

a. Weight	i. The force of the ground holding something up.
b. Reaction (type A)	ii. The force pushing something forward.
c. Reaction (type B)	iii. The force of water pushing upwards.
d. Thrust	iv. Gravity acting downwards on an object.
e. Friction	v. The force of an object pushing back against you.
f. Uplift	vi. The force acting against any movement.
g. Upthrust	vii. The force that lifts things up in air.

Task 2 – Adding Forces

Draw forces on these two objects.

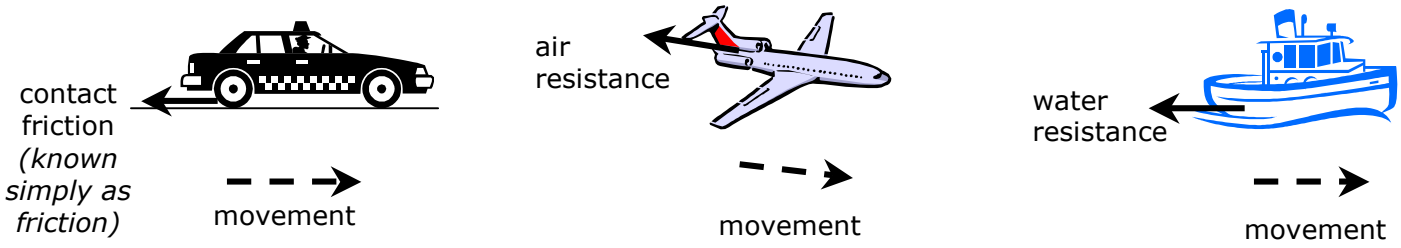




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Friction

Friction always acts **against** a moving object. Three different types of friction are shown in the pictures.



Task 3 – Types of Friction

Which type of friction acts on each of these objects?

a. A bicycle tyre _____

b. A fish _____

c. A firework rocket _____

d. A parachute _____

e. A jet ski _____

f. A sole of a shoe on the ground _____

g. A submarine _____

Task 4 - Forces on a parachutist

What type of force is being described in each of the statements below? Choose your answers from the forces shown in the previous tasks.

a. A plane is flying through the air. This force pulls the plane down towards the earth. _____

b. The wings of the plane have been designed to create this upward force. _____

c. The jet engines create this pushing force. _____

d. This force is caused by the air. It acts to slow the plane down. _____

e. A woman parachutist jumps out of the plane. This force pulls her towards the ground. _____

f. The woman opens her parachute. This force acts on the parachute to slow her down. _____

g. The woman lands on the ground. This is the force of the ground pushing against her weight. _____

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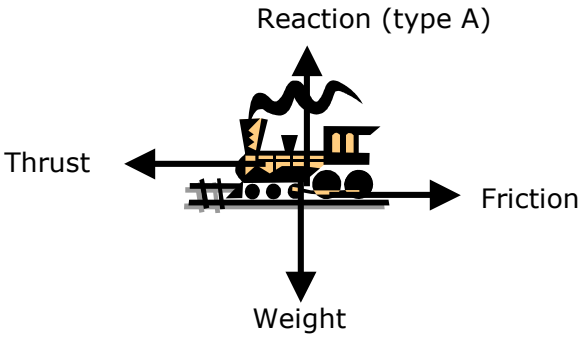
Types of Forces - Answers

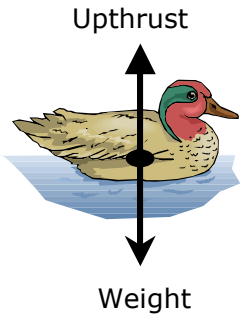
Forces 2

Task 1 - Force definitions

a.	Weight	**iv.**	Gravity acting downwards on an object.
b.	Reaction (type A)	**i.**	The force of the ground holding something up.
c.	Reaction (type B)	**v.**	The force of an object pushing back against you.
d.	Thrust	**ii.**	The force pushing something forward.
e.	Friction	**vi.**	The force acting against any movement.
f.	Uplift	**vii.**	The force that lifts things up in air.
g.	Upthrust	**iii.**	The force of water pushing upwards.

Task 2 – Adding Forces





Task 3 – Types of Friction

a. A bicycle tyre - *Friction*

b. A fish - *Water resistance*

c. A firework rocket - *Air resistance*

d. A parachute - *Air resistance*

e. A jet ski - *Water resistance and air resistance*

f. A sole of a shoe on the ground - *Friction*

g. A submarine - *Water resistance*

Task 4 - Forces on a parachutist

a. A plane is flying through the air. This force pulls the plane down towards the earth - *gravity*

b. The wings of the plane have been designed to create this upward force - *uplift*

c. The jet engines create this pushing force - *thrust*

d. This force is caused by the air. It acts to slow the plane down – *air resistance*

e. A woman parachutist jumps out of the plane. This force pulls her towards the ground - *gravity*

f. The woman opens her parachute. This force acts on the parachute to slow her down – *air resistance*

g. The woman lands on the ground. This is the force of the ground pushing against her weight – *reaction*

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